

Sequence similarity and phylogeny

Bioinformatic tools

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October 05, 2026



Quick poll: pe.app/cmaslard2

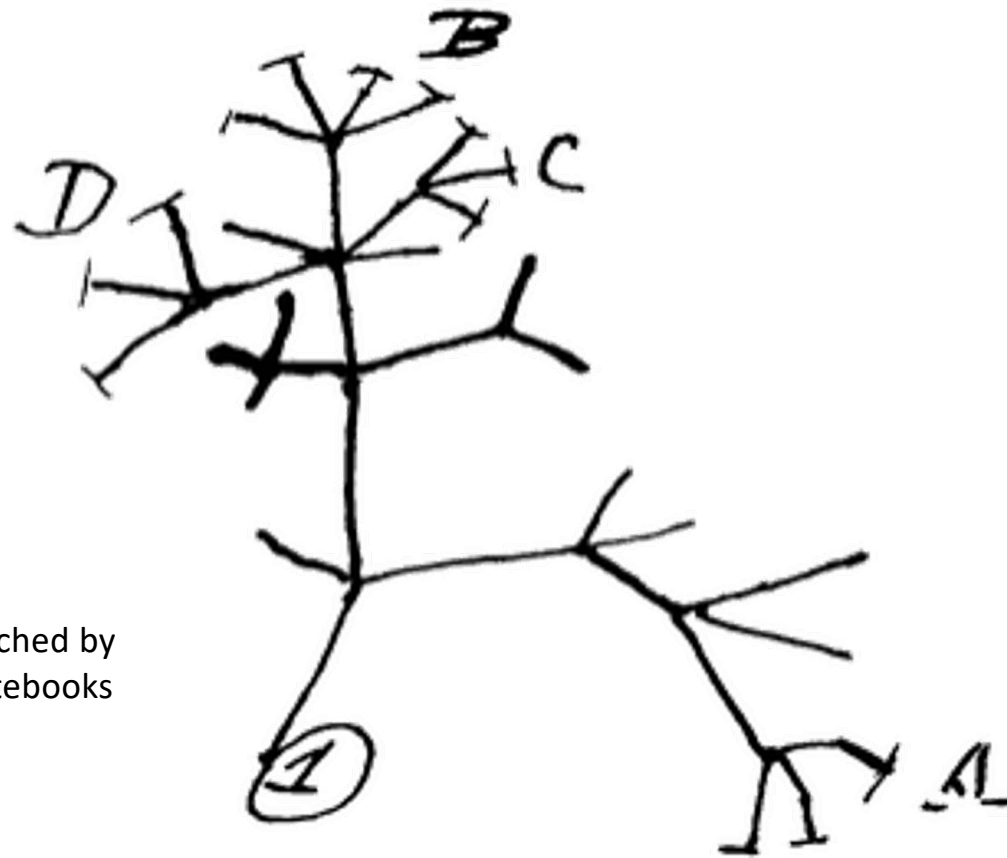
One question from Session 2 before we move on.

You are handed one unidentified sequence. Which of a handful of reference sequences is it most similar to, and what does that similarity tell you about where it came from?

Continuous assessment starts today

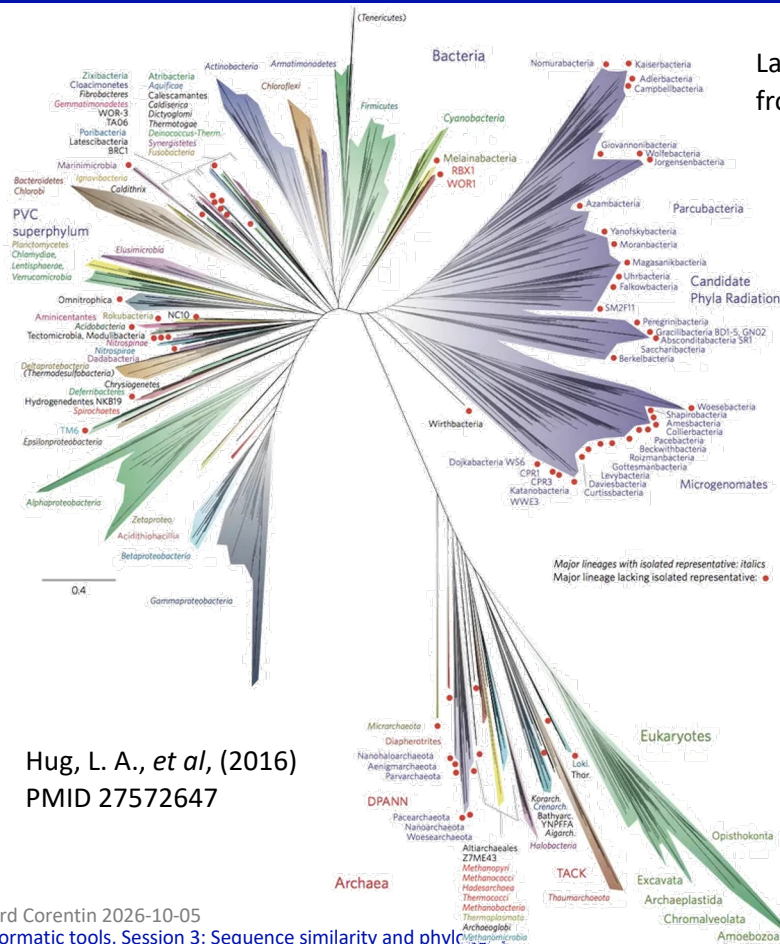
- Sessions 1-2 were not graded. **From today, every session ends with a short individual, graded task.**
- Today's required result: **one annotated tree, saved as an image, with a written biological interpretation.**
- Draft your interpretation in class, in Section 5 of WORKSHEET.md.
- Polish and submit by Thursday: [see the rubric at the end of today's practical.](#)

Phylogenetics and sequence analysis



The first evolutionary tree sketched by Darwin (1837) in one of his notebooks

Phylogenetics and sequence analysis



Late 19th century:
from Greek *phulon*, *phulē* 'race, tribe' + *genesis*.

The **evolutionary development** and diversification of a **species or group of organisms**, or of a particular feature of an organism.

A representation of the **hypothesized** evolutionary relationships among a group of organisms.

Similarity search: what BLAST measures

grep (Sessions 1-2) finds exact text. Biological sequences almost never match exactly: mutation, sequencing error, and evolutionary divergence all change a handful of bases here and there.

BLAST (Basic Local Alignment Search Tool) finds sequences that are similar enough, scores how similar, against thousands of sequences at once, in seconds.

Column	Meaning
percent identity	percent identity: of the aligned bases, what fraction are identical
length	how many bases the alignment covers
e-value	how many matches this good would be expected by chance alone; smaller is better
bitscore	overall match quality, normalized so it is comparable between searches; higher is better

A high percent identity over a short length can still be a coincidence. Block B gives you a real case where trusting percent identity alone would mislead you.

Where you have already seen this idea

- Session 1's mystery.fasta contained one record, sp|P0DTC2|SPIKE_SARS2, that was neither obviously DNA nor identifiable by eye.
- A similarity search is exactly the next step you would take on a record like that: not “does it look like DNA”, but “what does it match”.
- You meet that exact sequence again in today's two-minute UniProt detour.

From one match to a tree

BLAST answers “**what is my sequence most similar to**”, one hit at a time. A phylogenetic tree places every sequence on a single diagram of relative relatedness.

1. Multiple sequence alignment

Stack and pad every sequence with gaps so the same column means the same position in every sequence.

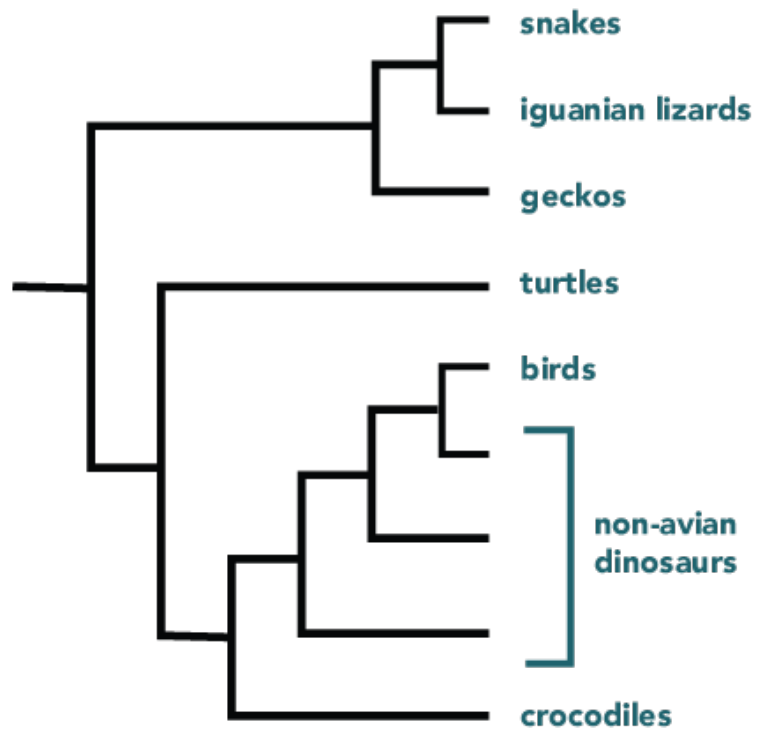
2. Distance

Count the fraction of columns that differ between every pair of sequences: a distance matrix.

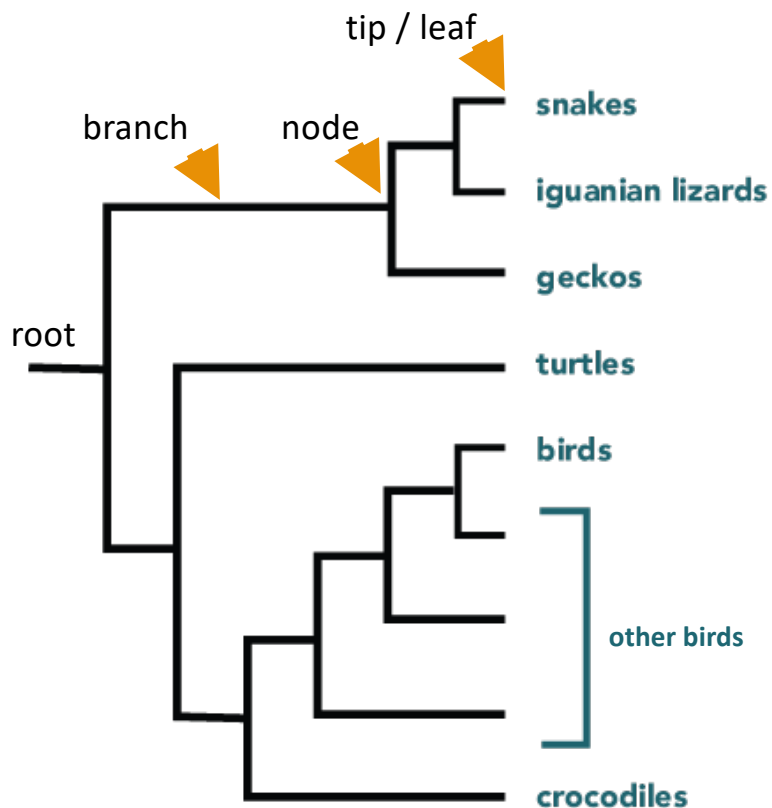
3. Tree building

Neighbour-joining arranges sequences so that a small distance sits close together, joined by a recent node.

How to describe it? / Or how to “read” a tree

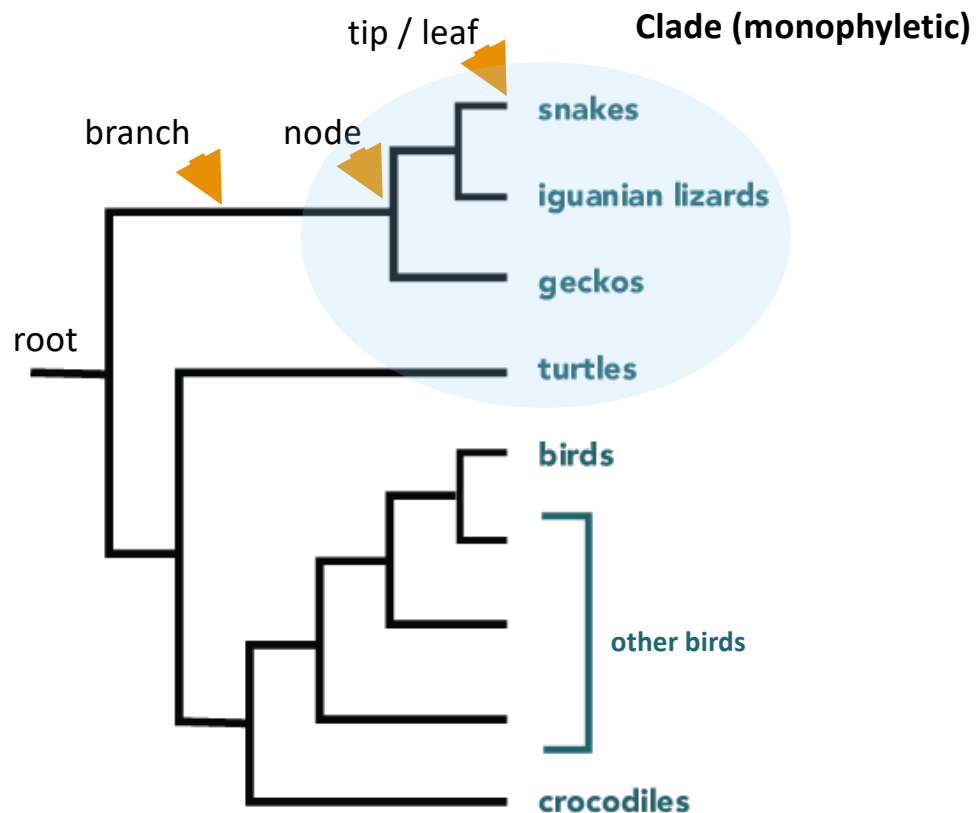


How to describe it? / Or how to “read” a tree



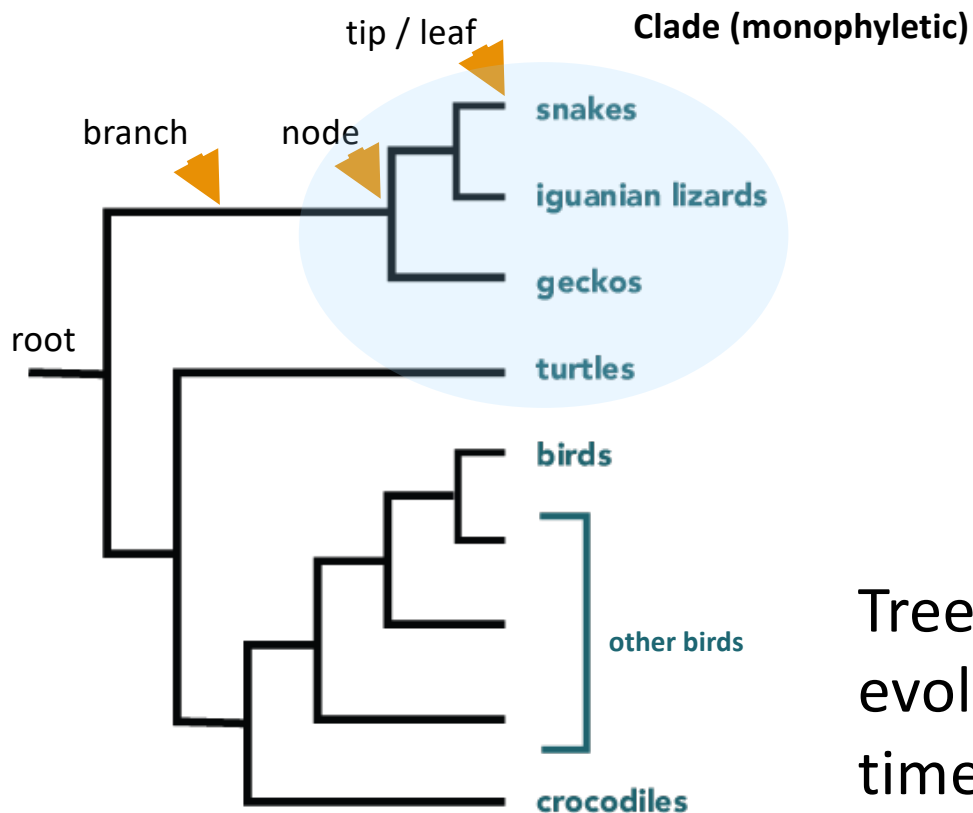
Term	Meaning
Tip	one sequence, labelled, at the end of a branch
Node	a split point: the inferred common ancestor of everything beyond it
Branch length	genetic distance between a node and what it leads to
Clade	a node plus everything descended from it
Outgroup	the most distantly related sequence, used to anchor the root

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What a phylogeny can be

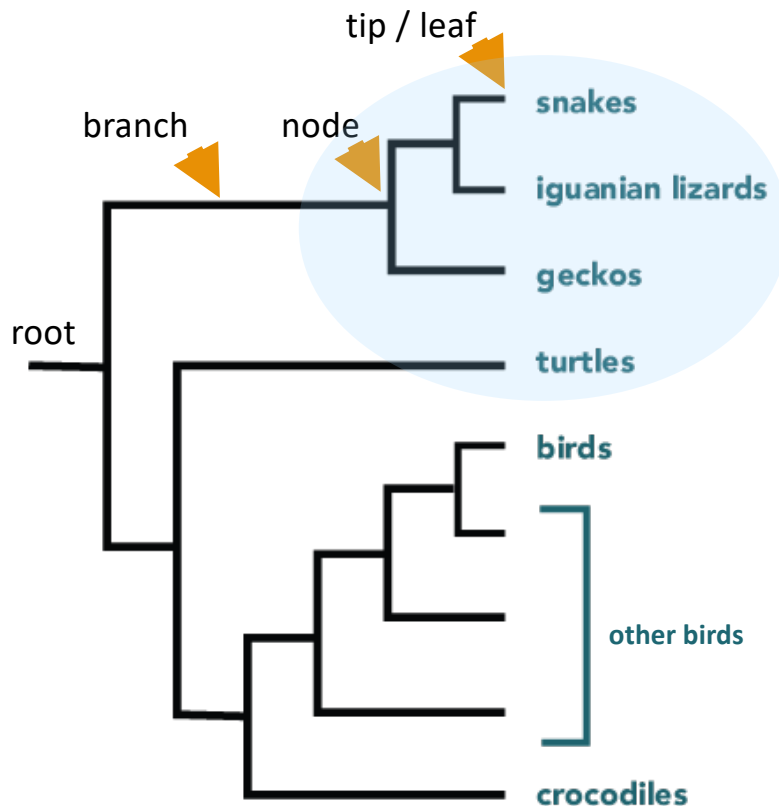


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Tree at different
evolutionary
timescales

i.e. inter or intraspecific, one
or multiple sequences
(species or gene trees)

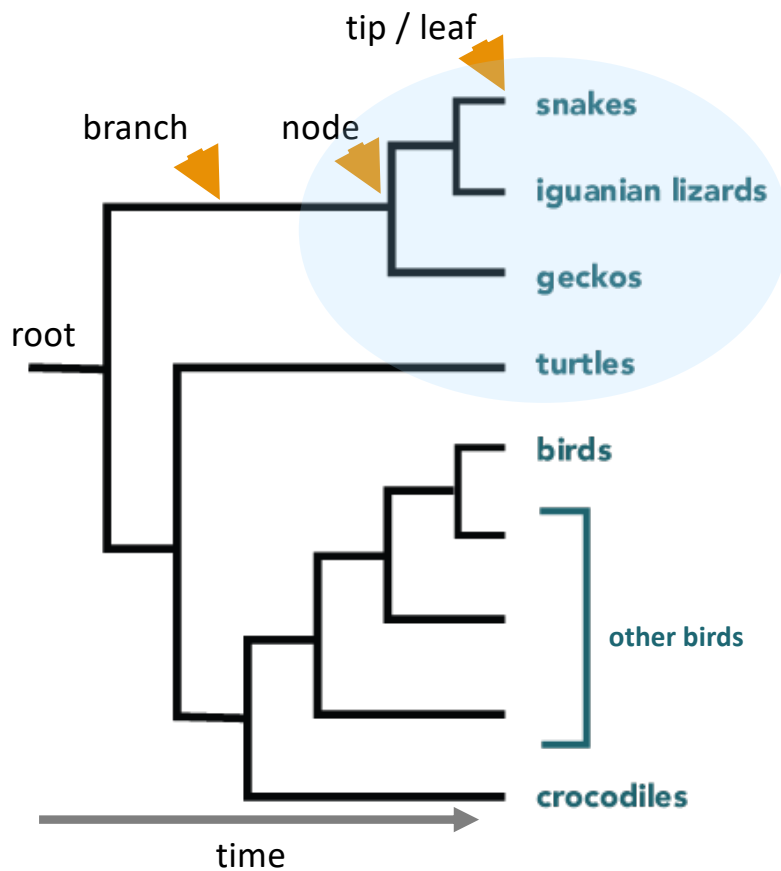
What a phylogeny can('t) tell



👍 Relationships among organisms or sequences (shared ancestry)

👎 Relationships among organisms or sequences (shared phenotype / function)

Common misconceptions



✗ No more “advanced” taxa

✗ Tip proximity = closely related

✗ % of nodes = relatedness

✗ Long branches = no change

How is related to who?

How did a trait evolve?

What is the origin of a disease?

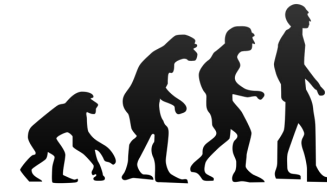
How is related to who?

- * Identify unfamiliar species has helped us understand the origins of many diseases.
- * Find relatives of endangered species can help us better conserve these species.



How did a trait evolve?

- * Find wild relatives of domestic species has helped us improve important crops.
- * Determine which species are likely to have inherited a trait has pointed the way towards new medications.

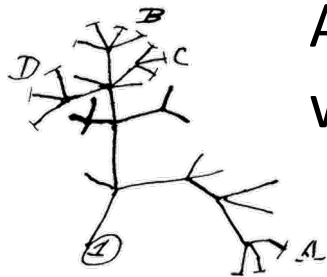


What is the origin of a disease?

- * Determine lineage-splitting events has helped uncover the true roots of diseases.
- * Evolutionary trees can change the way we look at the world around us.

Recently inherited sequences are more similar

I think



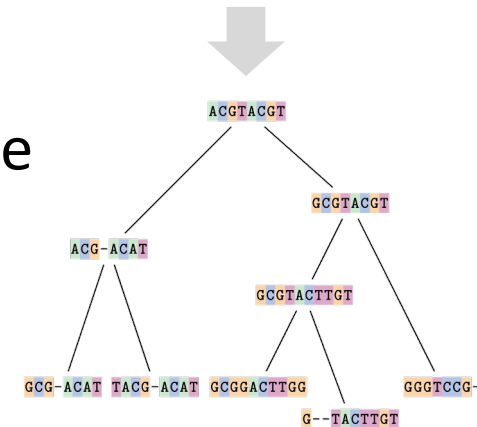
Although first phylogenies were based on traits

Limitations?



Multiple sequence alignment

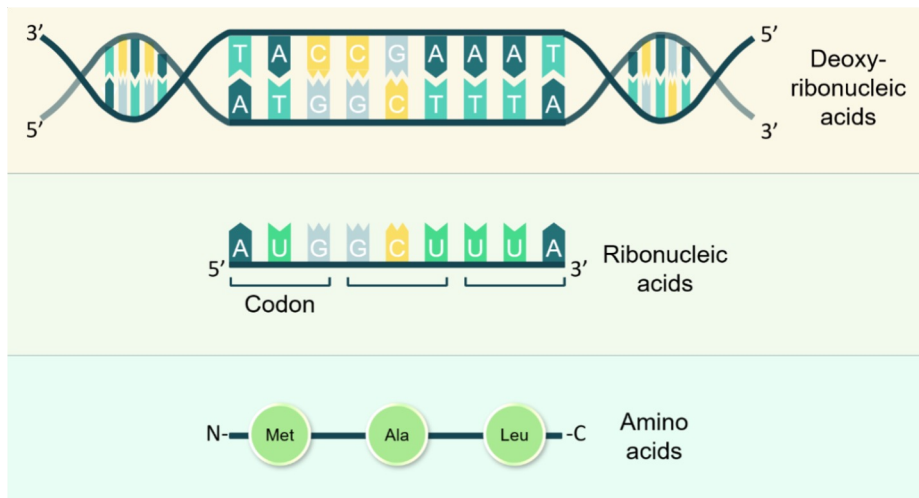
Phylogenetic trees are a representation of a multiple sequence alignment



How to infer relatedness?

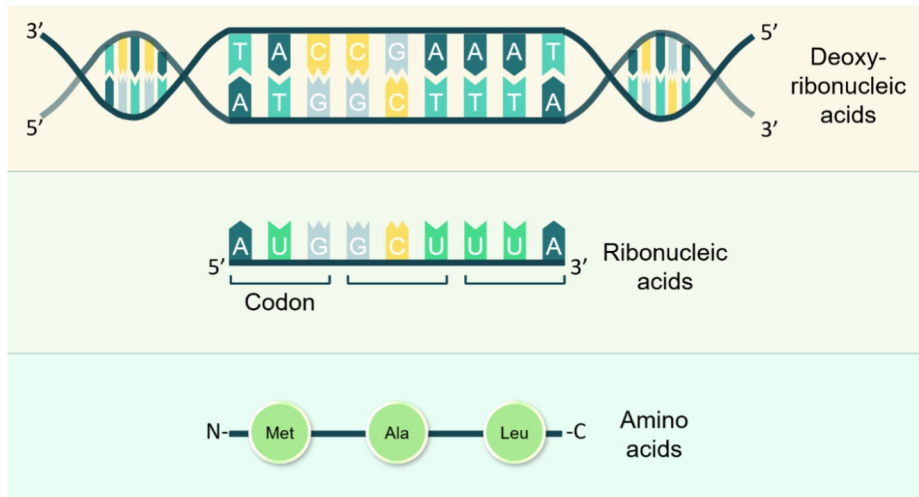
Or what do you need?

Different flavours of sequence



(i) sequences

Different flavours of sequence



(i) sequences

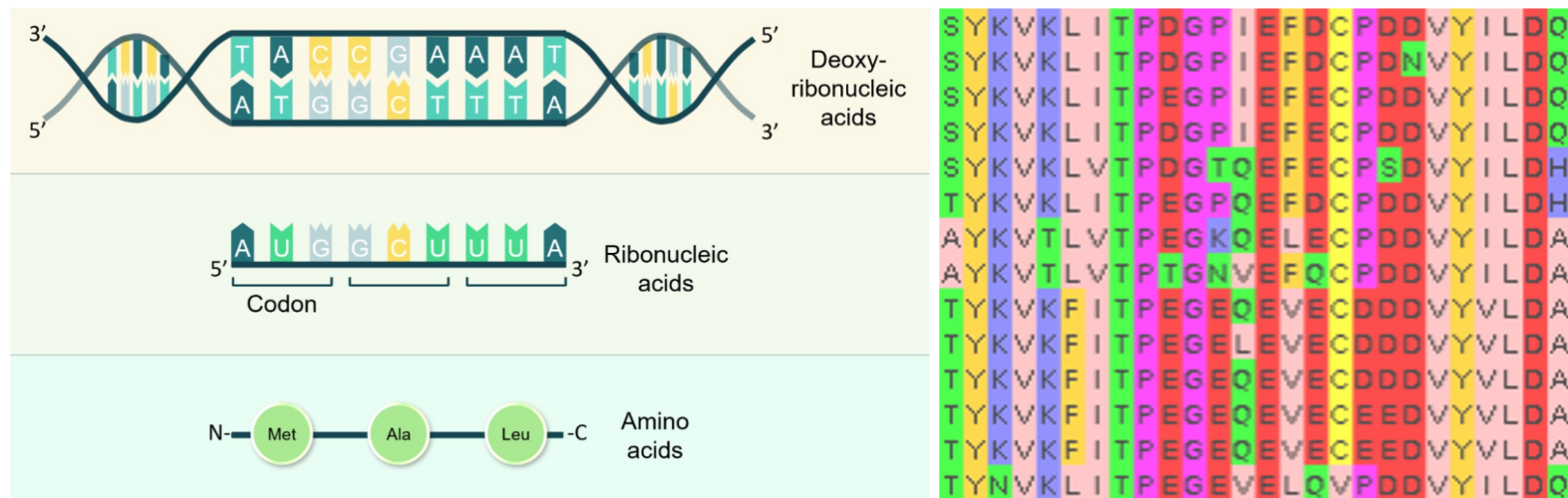
Easy to get – fast evolving – low divergence – high resolution

rRNA – conserved – deep divergence - low resolution

conserved – deep divergence - low resolution

Different flavours of sequence

(ii) tools (aligners)



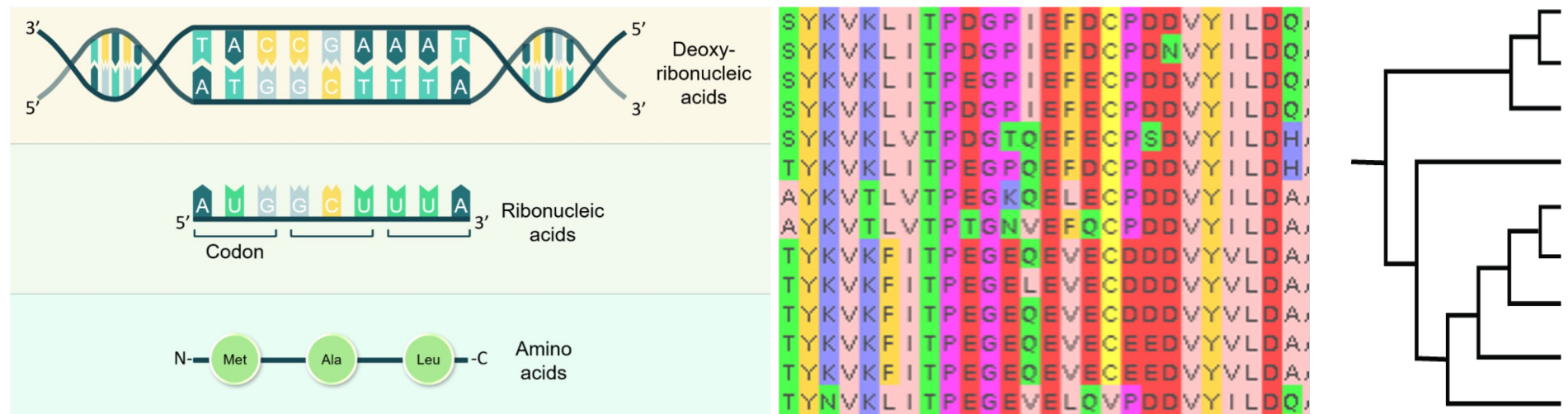
(i) sequences

Finding homologous residues

Method	How it works	Examples	Pros	Cons
Dynamic Programming (DP)	Builds scoring matrix (Needleman–Wunsch, Smith–Waterman); guarantees optimality.	Needleman–Wunsch, Smith–Waterman	Optimal solution; handles gaps & scoring matrices rigorously.	Computationally expensive ($O(n^2)$ time & space).
Heuristic (Database Search)	Uses k-mer “seeds” and extends hits to high-scoring pairs; skips full DP.	BLAST, FASTA, DIAMOND	Very fast; suitable for large databases; practical in genomics.	Approximate; may miss weak/distant homologies.
Progressive Alignment	Builds guide tree from pairwise scores, then aligns sequences stepwise.	ClustalW, MAFFT, MUSCLE	Fast, scalable to many sequences.	Early errors propagate; not optimal.
Iterative Alignment	Starts with an initial alignment, then repeatedly refines by realigning subsets.	MUSCLE, MAFFT (iterative mode), PRANK	Corrects earlier errors; generally more accurate than pure progressive.	More computationally intensive than progressive methods.
Consistency-based Alignment	Incorporates information from multiple pairwise alignments to improve MSA.	T-Coffee, ProbCons	Higher accuracy; reduces error propagation.	Slower; more computationally demanding.
Hidden Markov Models (HMMs)	Uses probabilistic models to represent sequence states (match, insert, delete).	HMMER, SAM, Infernal	Sensitive to remote homologies; robust statistical framework.	Requires training; slower than heuristics.
Profile–Profile Alignment	Aligns sequence profiles (MSA summaries) instead of raw sequences.	PSI-BLAST, HHsearch, HHPred	Detects distant relationships; improves sensitivity.	More complex; depends on quality of input profiles.
Structure-aware Alignment	Uses 3D structure information along with sequence data.	PROMALS3D, Expresso	Accurate for proteins with structural data.	Limited by availability of 3D structures.

Inferring relatedness

(ii) tools (aligners)



Inferring relatedness

Category	Methods	Assumption	Strengths	Weaknesses
Distance-based	UPGMA, Neighbor-Joining	Evolution summarized as pairwise distances (often using substitution models like JC, K2P).	Very fast; scalable; useful for exploratory trees.	Ignores site-specific information; can be misleading with unequal rates.
Parsimony (Character-based)	Maximum Parsimony (MP)	Best tree = fewest evolutionary changes; no explicit substitution model.	Simple, intuitive; works for morphology and discrete characters.	Sensitive to long-branch attraction; less accurate for molecular data.
Statistical (Character-based)	Maximum Likelihood (ML), Bayesian Inference (BI)	Explicit substitution models (JC, K80, GTR, codon models; rate heterogeneity via gamma).	Statistically rigorous; captures rate variation; BI gives posterior support.	Computationally intensive; BI especially slow; needs good model choice.
Multi-locus / Coalescent	ASTRAL, *BEAST, species tree methods	Gene trees evolve within species tree under population genetics (coalescent).	Handles incomplete lineage sorting; essential for genome-scale datasets.	Requires many loci; assumptions more complex; heavy computation.

Your Q defines the data and method

Multiple Sequence Alignment (MSA)

Few sequences

Dynamic Programming — Needleman–Wunsch, Smith–Waterman

Many sequences

Fast & scalable → **Progressive** — ClustalW, MAFFT, MUSCLE

Improve accuracy by re-aligning → **Iterative** — MUSCLE, MAFFT

Prevent early errors using pairwise agreement → **Consistency-based** - T-Coffee, ProbCons

Phylogenetic Inference

Small dataset

Morphology / discrete characters → **Maximum Parsimony** (MP)

Molecular data

Need speed / exploratory tree → **Distance-based** (Neighbor-Joining, UPGMA)

Need accuracy / model-based → **Maximum Likelihood** (ML)

Large dataset

Single locus → **ML** (IQ-TREE, RAxML, PhyML)

Multi-locus / genome-scale

Ignore gene-tree discordance → **Concatenation + ML** (supermatrix)

Model gene-tree discordance → **Coalescent-based methods** (ASTRAL, *BEAST)

Worked example: a family of coronaviruses

Worked example: a family of coronaviruses

SARS-CoV-2, SARS-CoV-1 (the 2003 SARS outbreak), MERS-CoV, and several seasonal “common cold” coronaviruses are all related, but to very different degrees.

Comparing one gene, the spike gene (S), the gene for the protein coronaviruses use to enter a host cell, is how virologists work out which viruses are close relatives and which are more like distant cousins.

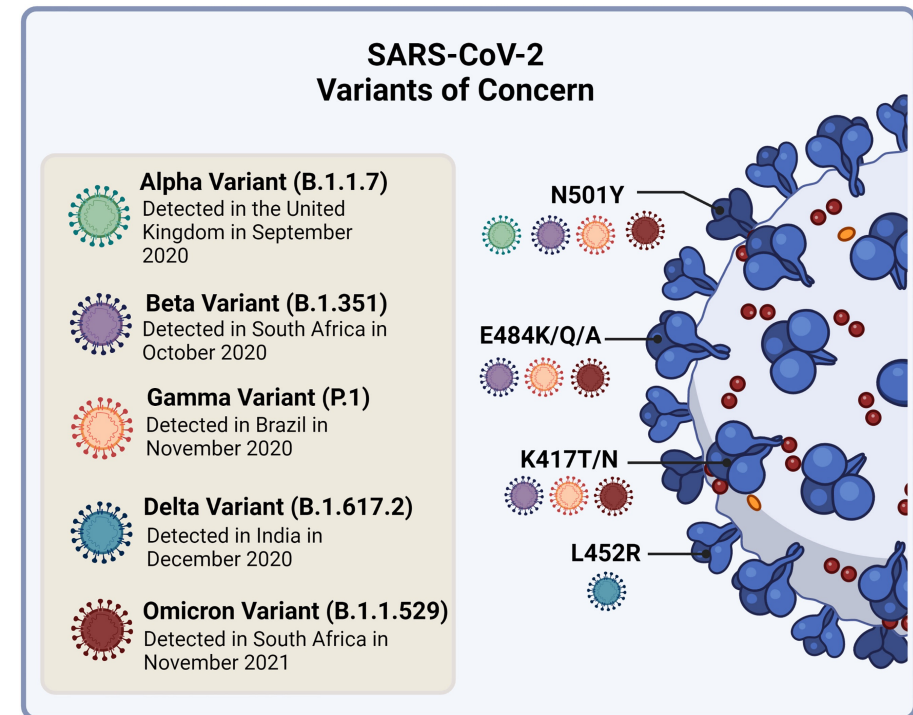
Today: 1 unidentified sequence + 4 named references, all spike genes, from different coronaviruses - checked at both the nucleotide and the protein level.

The New York Times

2020

China Grapples With Mystery Pneumonia-Like Illness

Beijing is racing to identify a new illness that has sickened 59 people as it tries to calm a nervous public.



p. 29

SARS-CoV-2 Variants of Concern



Alpha Variant (B.1.1.7)

Detected in the United Kingdom in September 2020



Beta Variant (B.1.351)

Detected in South Africa in October 2020



Gamma Variant (P.1)

Detected in Brazil in November 2020



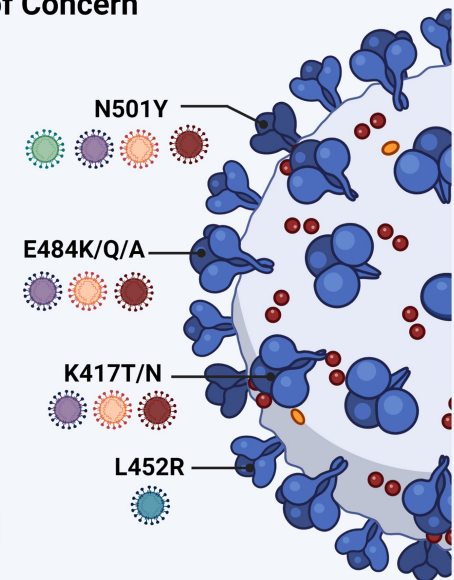
Delta Variant (B.1.617.2)

Detected in India in December 2020



Omicron Variant (B.1.1.529)

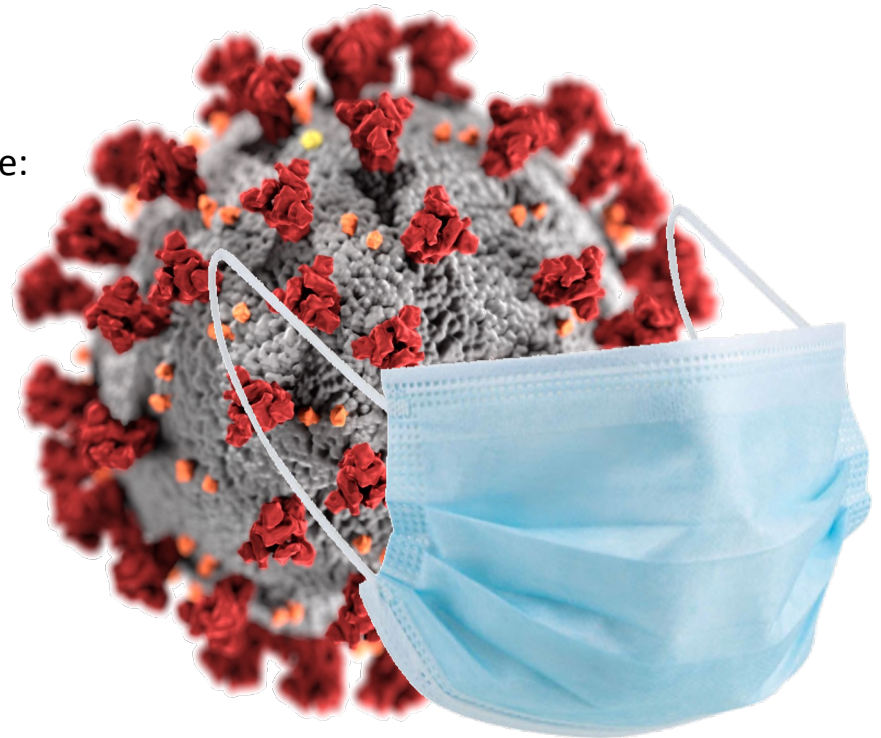
Detected in South Africa in November 2021



<https://doi.org/10.3389/fimmu.2021.809244>

Two-minute detour: the same logic, via UniProt

- Session 1's unidentified record, sp|P0DTC2|SPIKE_SARS2, is the SARS-CoV-2 spike protein.
- P0DTC2 is its accession in UniProt, a curated protein database: look it up and, instead of raw sequence, get functional annotation - named domains (receptor-binding domain, furin cleavage site, signal peptide, transmembrane region), and links to solved 3D structures.
- A protein search (blastp) works exactly like blastn, just on amino acids instead of bases. Block D2 has you run it on this exact protein.



Practical briefing: what's ahead (15:00-17:00)

1. Fetch today's sequences: one unknown, four named references.
2. Two BLAST searches against the references: quick, then deep.
3. Inspect a prepared alignment and distance matrix by hand.
4. Render and annotate a phylogenetic tree in R (pairs).
5. Repeat the comparison at the protein level (pairs).
6. Draft your interpretation; polish and submit by Thursday.

Toolbox for today

New:

`module load BLAST+, makeblastdb, blastn, blastp, source()`

Still yours from Sessions 1-2:

`cd, ls, mkdir, cp, head, grep, wc -l, module load`